

CS520 Project Requirements

Coverd: UMass Shift Management System

Team 17

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Link to project document:  CS520 Project Requirements – Shift Management System

Link to public github repository: <https://github.com/COMPSCI-520/coverd>

1. Requirements

1.1 Overview

Student workers at the UMass dining commons currently rely on manually maintained Google Sheets to manage their work schedules. Many student employees, including a large number of international students, work multiple shifts per week ranging from three to eight hours. The current scheduling process creates several issues. Personal availability and schedules are visible to all staff members, raising privacy concerns. In addition, the process of dropping a shift or finding a replacement is highly disorganized, often relying on scattered communication channels such as group messages or personal texts. Managers must manually track these requests, making it difficult to maintain accurate records of schedule changes.

Another major concern involves compliance with the legal work restrictions for international students, who are limited to a maximum of 20 work hours per week during the academic semester. Without an automated system to track cumulative work hours, students may accidentally exceed the allowed limit when picking up additional shifts from peers. To address these issues, this project proposes a centralized Shift Management System designed specifically for student workers and dining commons managers within the Five College community. The system will allow students to view their schedules, request shift drops, and pick up available shifts through a centralized platform, while managers can efficiently monitor and approve schedule changes through an administrative dashboard.

1.2 Features

1. User Authentication and Role-Based Access

- Secure login system for student workers and dining managers.

2. Shift Schedule Viewing

- Students can view their assigned shifts and total weekly working hours.

3. Shift Drop and Shift Marketplace

- Students can drop shifts and browse available shifts posted by other workers.

4. Manager Approval Dashboard

- Managers can review, approve, or deny shift drop and shift swap requests.

5. Work Hour Compliance Tracking

- The system automatically calculates total weekly hours and prevents students from exceeding the 20-hour legal work limit.
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1.3 Functional Requirements (Use Cases)

Use Case 1: Student Login

Actor: Student Worker

Page: Login Page

Trigger: Student enters login credentials

Main Success Scenario

1. Student navigates to the Login Page.
2. Student enters username and password.

3. System verifies credentials.
4. System redirects the student to the Student Dashboard.

Failure Cases

- Invalid username or password.
 - User account not registered.
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Use Case 2: View Weekly Schedule

Actor: Student Worker

Page: Student Dashboard

Trigger: Student opens dashboard after login

Main Success Scenario

1. Student navigates to the Student Dashboard.
2. System retrieves scheduled shifts from the database.
3. System displays the weekly schedule.
4. System displays the student's total accumulated hours for the week.

Failure Cases

- Schedule data unavailable.
 - Database connection failure.
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Use Case 3: Request to Drop a Shift

Actor: Student Worker

Page: My Schedule Page

Trigger: Student selects an assigned shift and clicks “Drop Shift”

Main Success Scenario

1. Student navigates to the My Schedule Page.
2. Student selects the shift to drop.
3. Student submits a drop request.
4. System records the drop request.
5. Manager receives notification.
6. The shift becomes available in the Shift Marketplace if the manager approves.

Failure Cases

- Shift already reassigned.
 - Drop request submitted after allowed deadline.
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Use Case 4: Browse Available Shifts

Actor: Student Worker

Page: Shift Marketplace Page

Trigger: Student opens the shift marketplace

Main Success Scenario

1. Student navigates to the Shift Marketplace Page.
2. System retrieves available shifts.
3. System displays shift details including time, location, and hours.
4. Student can select a shift to claim.

Failure Cases

- No shifts available.
 - System unable to retrieve shift data.
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Use Case 5: Claim an Available Shift

Actor: Student Worker

Page: Shift Marketplace Page

Trigger: Student clicks "Claim Shift"

Main Success Scenario

1. Student selects a shift from the marketplace.
2. System calculates the student's cumulative weekly hours.
3. If total hours remain ≤ 20 , the system accepts the claim request.
4. Manager receives approval request.

Failure Cases

- Claim would exceed the 20-hour weekly limit.
 - Shift already claimed by another student.
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Use Case 6: Manager Reviews Shift Requests

Actor: Manager

Page: Manager Dashboard

Trigger: Manager opens pending requests

Main Success Scenario

1. Manager navigates to the Manager Dashboard.
2. Manager views pending shift requests.
3. Manager selects a request.
4. Manager approves or rejects the request.
5. System updates the shift schedule accordingly.

Failure Cases

- Request no longer valid.
 - Manager cancels the action.
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Use Case 7: Automatic Hour Limit Enforcement

Actor: System

Page: Shift Marketplace Page

Trigger: Student attempts to claim a shift

Main Success Scenario

1. Student attempts to claim a shift.
2. System calculates the student's total weekly hours.
3. System verifies the total does not exceed 20 hours.
4. If compliant, the request proceeds.

Failure Cases

- Hour limit exceeded.
- System blocks the request and displays a warning message.

Use Case 8: Manager Views Staff Schedule

Actor: Manager

Page: Staff Schedule Page (Manager Interface)

Trigger: Manager opens schedule overview

Main Success Scenario

1. Manager navigates to the Staff Schedule Page.
2. Manager selects a date or week range.
3. System retrieves schedules for all student workers.
4. System displays assigned shifts.

Failure Cases

- Schedule data unavailable.
- Database retrieval error.

1.4 Non-Functional Requirements

Performance

- The system shall support at least **200 concurrent users** without noticeable performance degradation.

Security

- All user authentication data shall be securely hashed and protected during storage and transmission.

Compliance Enforcement

- The system shall automatically enforce the **20-hour weekly work limit** for international students.

Availability

- The system shall maintain **99% uptime during operational hours**.
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1.5 Challenges & Risks

Risk 1: Concurrent Shift Claims

Multiple students may attempt to claim the same shift at the same time.

Mitigation

- Implement database locking or atomic transactions.
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Risk 2: Compliance Calculation Errors

Incorrect hour calculations could allow students to exceed legal limits.

Mitigation

- Implement automated validation and unit tests for hour calculations.
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Risk 3: Performance During Peak Usage

Many students may access the system simultaneously when shifts become available.

Mitigation

- Conduct load testing and optimize database queries.